🌐 Lab: VLAN Logical Segmentation & Layer 2 Security

📌 Project Overview

This lab demonstrates the implementation of **Virtual Local Area Networks (VLANs)** on a Cisco 2960 Switch using **Cisco Packet Tracer**. The goal was to take a flat network and segment it into three distinct departments (IT, HR, and Finance) to improve security and reduce broadcast traffic.

🎯 Objective

- Design a network where three departments share a single physical switch but remain logically isolated.
- Configure access ports and assign them to specific VLAN IDs.
- Verify that devices on the same subnet but different VLANs cannot communicate (Layer 2 Isolation).

🛠️ Technical Setup

- **Software:** Cisco Packet Tracer
- **Hardware:** 1x Cisco Catalyst 2960 Switch
- **End Devices:** 3x Generic PCs
- **Addressing:** All devices reside on the `192.168.1.0/24` subnet.

Department	VLAN ID	Port Assignment	IP Address
IT	10	Fa0/1	192.168.1.1 0
HR	20	Fa0/2	192.168.1.11
Finance	30	Fa0/3	192.168.1.1

Export to Sheets

💻 Configuration (Cisco IOS)

The following commands were used to initialize the VLAN database and assign physical interfaces:

```
Bash
# Creating the VLANs
Switch(config)# vlan 10
Switch(config-vlan)# name IT
Switch(config)# vlan 20
Switch(config-vlan)# name HR
Switch(config)# vlan 30
Switch(config-vlan)# name FINANCE

# Assigning interfaces to Access Mode
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
```

🔍 Verification & Testing

1. **VLAN Database Check:** Ran `show vlan brief` to confirm ports were correctly mapped to their respective IDs.
2. **Connectivity Test:** Attempted a `ping` from the IT PC to the HR PC.
    - **Expected Result:** Request Timed Out.
    - **Actual Result:** Request Timed Out.
3. **Conclusion:** The switch successfully dropped the frames because they belonged to different broadcast domains, confirming the security of the logical segmentation.

🧠 Skills Demonstrated

- Cisco IOS Command Line Interface (CLI)
- Layer 2 Switching & Segmentation
- Broadcast Domain Management
- Network Troubleshooting & ICMP Testing

Command Prompt

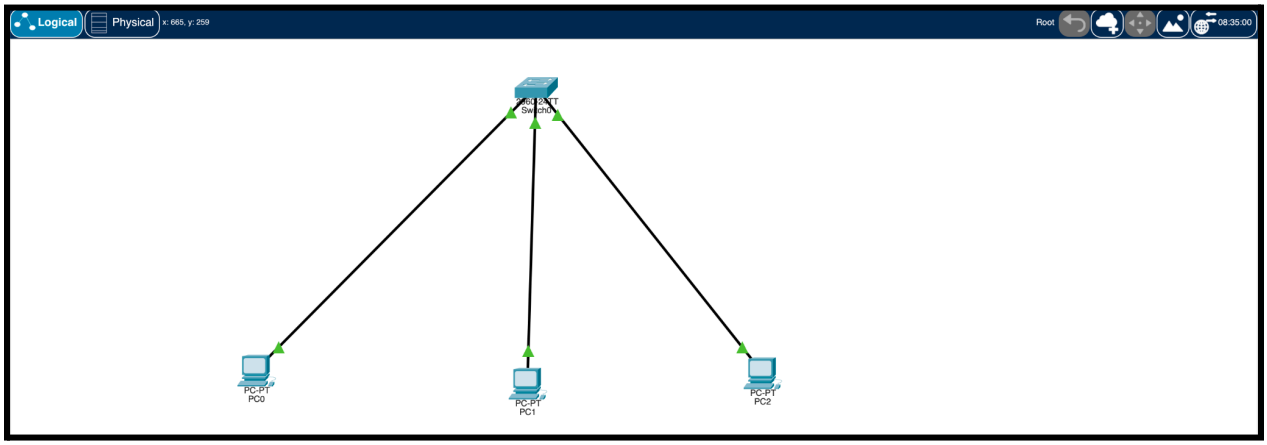
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=16ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 16ms, Average = 4ms
```

IP Configuration

Interface	FastEthernet0

IP Configuration

DHCP	Static
IPv4 Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0



```
Switch>ENABLE
Switch#enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)#name IT
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name FINANCE
Switch(config-vlan)#
Switch(config-vlan)#ecit
                     ^
% Invalid input detected at '^' marker.

Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#show vlan
                ^
% Invalid input detected at '^' marker.

Switch(config)#
Switch(config)#
Switch(config)#HOSTNAME SW1
SW1(config)#
SW1(config)#enable secret yasin
SW1(config)#exit
```

```
SW1#confi
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#
SW1(config)#interface f0/1
SW1(config-if)#
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#
SW1(config-if)#
SW1(config-if)#interface f0/2
SW1(config-if)#switchport mode access
SW1(config-if)#switchport acess vlan 20
                            ^
% Invalid input detected at '^' marker.

SW1(config-if)#interface f0/2
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 20
SW1(config-if)#
SW1(config-if)#
SW1(config-if)#interface f0/3
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 30
SW1(config-if)#
SW1(config-if)#exit
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show vlan bre
SW1#show vlan bre
               ^
% Invalid input detected at '^' marker.
```

```
% Invalid input detected at '^' marker.

SW1#
SW1#show vlan bri
SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   IT                               active    Fa0/1
20   HR                               active    Fa0/2
30   FINANCE                          active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1#
SW1#
```

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=16ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 16ms, Average = 4ms

C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

```
Switch>ENABLE
Switch#enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)#name IT
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name FINANCE
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#show vlan

SW1#configure terminal
```

```
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#
SW1(config)#line console 0
SW1(config-line)#password ahmed
SW1(config-line)#login
SW1(config-line)#
SW1(config-line)#exit
SW1(config)#
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console
SW1#exit
SW1 con0 is now available
Press RETURN to get started.
User Access Verification
Password:
SW1>enable
Password:
SW1#
SW1#confi
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#
SW1(config)#interface f0/1
SW1(config-if)#
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#
SW1(config-if)#
SW1(config-if)#interface f0/2
SW1(config-if)#switchport mode access
SW1(config-if)#switchport acess vlan 20
                      ^
% Invalid input detected at '^' marker.

SW1(config-if)#interface f0/2
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 20
SW1(config-if)#
SW1(config-if)#
SW1(config-if)#interface f0/3
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 30
SW1(config-if)#
SW1(config-if)#exit
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console
```

```
SW1#show vlan bre
SW1#show vlan bre
            ^
% Invalid input detected at '^' marker.

SW1#
SW1#show vlan bri
SW1#show vlan brief
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   IT                               active    Fa0/1
20   HR                               active    Fa0/2
30   FINANCE                          active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1#
SW1#
```